

ST-GCN — Spatial-Temporal Graph Convolutional Network

Per-model deep dive · the anatomical prior, and the low-variance workhorse

Result: 0.448 accuracy · 0.389 weighted F1 · WER 0.552 · fold SD ± 0.059

Notebook: 04_stgcn_wlasl_landmark_baseline.ipynb

1 Theory

An ST-GCN treats each frame not as a flat vector but as a **skeleton graph**: joints are nodes, and the body's physical connections (wrist–elbow, finger joints) are edges. It convolves over that graph in space (a joint's neighbours inform it) and over time (a temporal convolution across frames). The inductive bias is anatomical: connected joints move together, and this structure is given to the model rather than learned from scratch — a powerful prior when data is scarce.

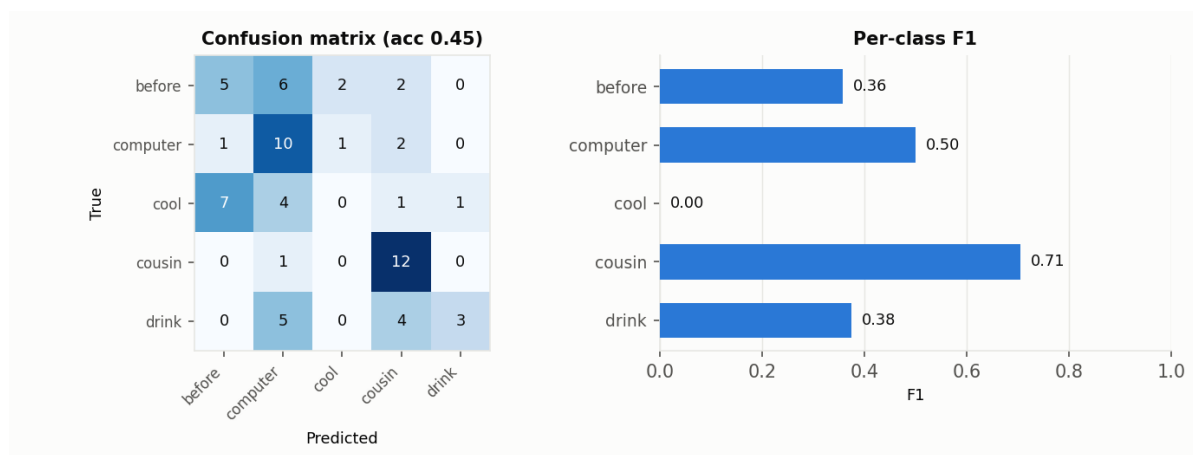
2 Method of execution

Each frame is a 75-node graph (33 pose + 21 + 21 hand joints; face dropped) with 3 coordinates per node. A hand-built adjacency matrix encodes the MediaPipe pose and hand edges, adds self-loops, and is symmetrically normalized. Stacked ST-GCN blocks each perform a spatial graph convolution (einsum over the adjacency) followed by a temporal convolution, with BatchNorm, ReLU, dropout and a residual path; a global average pool feeds a linear classifier. The graph convolution is written directly — no external graph-learning library.

```
A = normalize(pose_edges + hand_edges + self_loops)      # 75x75
block: x → einsum('nctv,vw->nctw', x, A) → 1x1 conv → temporal conv(k) → BN/ReLU/res
3 blocks (3→32→64→64) → global pool → linear → 5 classes
```

3 Results

Metric	Value (5-fold CV)
Accuracy	0.448
Weighted precision	0.412
Weighted recall	0.448
Weighted F1	0.389
Word Error Rate (WER)	0.552
Fold accuracies	0.36, 0.43, 0.54, 0.46, 0.46
Cross-val SD	± 0.059
Distinct classes predicted	5 of 5
Majority baseline	0.224



ST-GCN confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

ST-GCN reaches 0.448 with the **lowest fold variance of any deep model** (SD ± 0.059). That stability is the story: the skeleton graph is a correct, strong prior, so the network does not have to discover body structure from 54 videos — it is told. That keeps learning well-behaved where the LSTM and Transformer swing wildly across folds. Its accuracy ceiling is held down here by two things: the face (and its mouthing cues) is dropped from the graph, and the sample is tiny. It is the most *reliable* deep model even though it is not the most accurate single one.

What makes it stronger / weaker

Stronger: anatomically grounded, parameter-efficient, and the most stable deep model at this data scale — exactly the property you want in a component you will fuse. **Weaker:** the fixed graph can't learn long-range frame-to-frame relations the way attention can, and dropping the face loses facial-expression signal that matters for some signs.

5 Role in the hybrid — a core component

ST-GCN is **one of the two streams in the recommended hybrid**. It contributes the stable, low-variance anatomical representation, while the Transformer stream contributes temporal expressiveness. Fusing their features gives the hybrid ST-GCN's reliability together with the Transformer's ceiling — which is why the two-stream model tops the study. In short: ST-GCN is the hybrid's **backbone of stability**.